

Well Completion Report: Offset Loss Incident Well MN-DW-03 (illustrative)

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1. Executive Summary

Drilled 2.4 km south-southeast of MN-SM-DW-01. At depth 4,222 m MD, well experienced severe mud losses (28 bbl/hr) following an unmanaged drilling break (ROP 26 m/hr), where annular ECD reached **12.18 ppg**, exceeding the shoe fracture integrity limit (12.10 ppg).

2. Sequence of Events

- Mud weight was elevated to 11.70 ppg to protect against sand overpressure.
- At 4,222 m MD, the bit entered a highly porous sand streak; ROP doubled from 13 m/hr to 26 m/hr without driller intervention.
- Cuttings loading in the 12-1/4 in annulus caused downhole ECD to spike to 12.18 ppg.
- The 13-3/8 in casing shoe at 3,860 m broke down, sustaining 28 bbl/hr continuous loss.
- Total mud lost to formation: 310 bbl before LCM pill (calcium carbonate blend 40 ppb) restored partial integrity.

3. Crucial Lessons

- In ultra-narrow operating margins (11.65 ppg PP vs 12.10 ppg FIT), drilling breaks must be immediately choked by imposing an ROP limit of $\leq 12 \text{ m/hr}$.